INCH-POUND

MIL-DTL-17/119G
w/AMENDMENT 1
13 April 2010

SUPERSEDING
MIL-C-17/119G
18 July 1985

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/119-RG174

Inactive for new design after 13 August 1993. For new design use MIL-C-17/196.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

NOTE: This cable uses PVC material and is not to be used in enclosed environments or shipboard applications. The replacements that are to be used in enclosed areas or shipboard applications are referenced in the following table.

The Air Force has restricted use of PVC in aerospace and ground support applications.

Cables listed on the current QPL may continue to be manufactured and supplied for existing enclosed applications only for a period not to exceed 3 years from the date of this specification.

TABLE I. Cross-reference data.

Part or Identifying Number (PIN)	Replacement PIN
M17/119-RG174	M17/196-00001

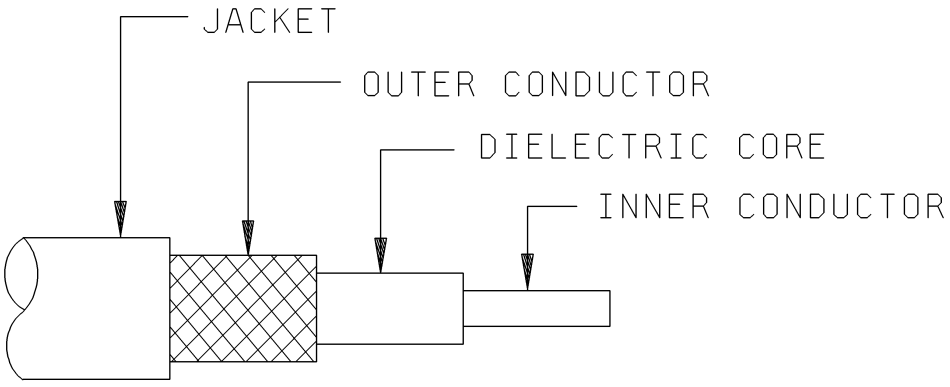


FIGURE 1. Configuration.

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TABLE II. Description.

Components	Construction details
Inner conductor	Seven strands of copper-covered steel wire, each strand .0063 inch diameter. Overall diameter: 0.0189 inch each ± 0.0010.
Dielectric core	Type A-1: Solid, polyethylene Diameter: 0.060 inch ± 0.003.
Outer conductor	Single braid of AWG #38, tinned copper wire. Diameter: 0.088 inch maximum. Coverage: 85.6% nominal Carriers: 16 Ends: 4 Picks/inch: 16.3 ± 10%
Jacket	Type IIa: PVC. Diameter: 0.110 inch ± 0.005.

ENGINEERING INFORMATION

Continuous working voltage: 1,100 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40C° to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 9.67 ohms per 100 feet.

Elongation: 1 percent, minimum.

Tensile strength: 110klb$_f$/inch^2 , minimum.

Engineering notes: This cable useful in general purpose, medium low temperature applications. (See connector series “SMA” and “SMB” in accordance with MIL-PRF-39012.)

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REQUIREMENTS

Dimensions, configuration, and description: See figure 1 and table II.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 4 pounds, minimum; 12 pounds, maximum.

Aging stability: +98ºC ± 2ºC.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend: -55ºC ± 2ºC.

Dimensional stability: +85C° ± 2°C.

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 0.0095 pound per foot, maximum.

Electrical:

Continuity: Applicable.

Spark test: 2,000 V rms, +10%, -0%.

Voltage withstanding: 2,000 V rms, +10%, -0%

Insulation resistance: Not applicable.

Corona extinction voltage: 1,500 V rms, minimum.

Characteristic impedance: 50 ohms ± 2.

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Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

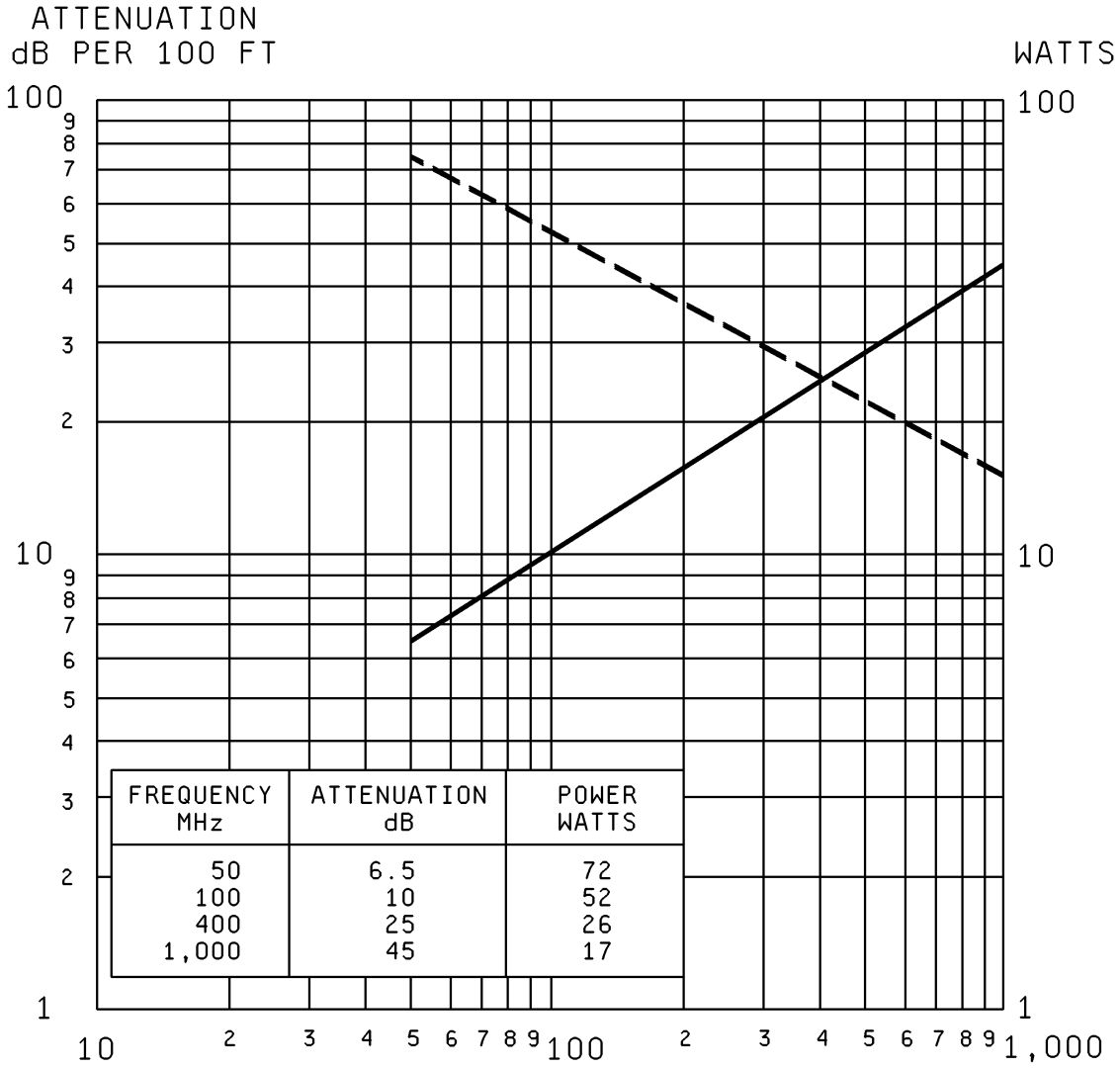
Transmission unbalance: Not applicable.

Mechanically induced noise: Not applicable.

Time delay: Not applicable.

PIN: M17/119-RG194 (See table I.)

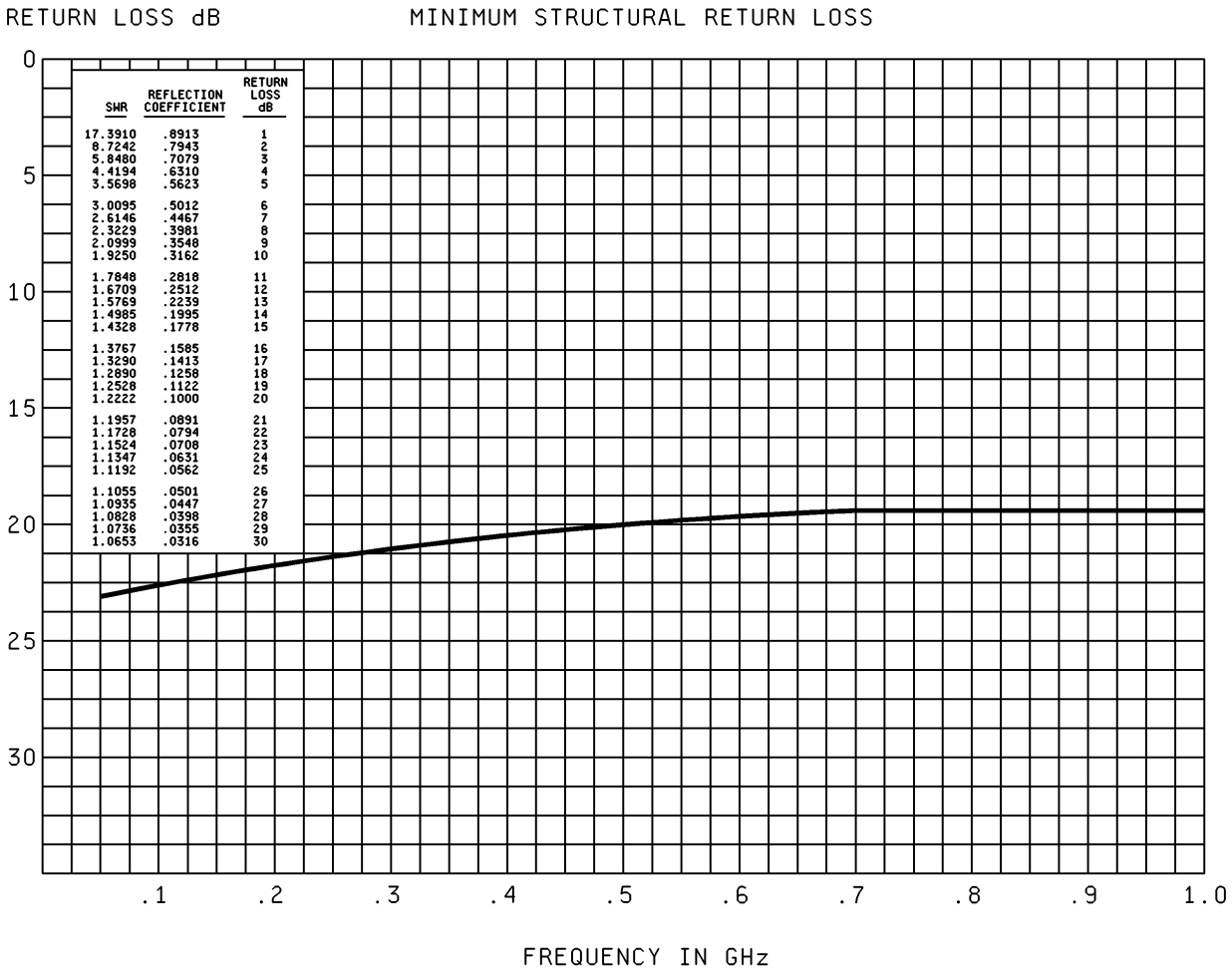
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FREQUENCY MHz	ATTENUATION dB	POWER WATTS
50	6.5	72
100	10	52
400	25	26
1,000	45	17

FIGURE 2. Power rating and attenuation.

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SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.3910	.8913	1
8.7242	.7943	2
5.8480	.7079	3
4.4194	.6310	4
3.5698	.5623	5
3.0095	.5012	6
2.6146	.4467	7
2.3229	.3981	8
2.0999	.3548	9
1.9250	.3162	10
1.7848	.2818	11
1.6709	.2512	12
1.5769	.2239	13
1.4985	.1995	14
1.4328	.1778	15
1.3767	.1585	16
1.3290	.1413	17
1.2890	.1258	18
1.2528	.1122	19
1.2222	.1000	20
1.1957	.0891	21
1.1728	.0794	22
1.1524	.0708	23
1.1347	.0631	24
1.1192	.0562	25
1.1055	.0501	26
1.0935	.0447	27
1.0828	.0398	28
1.0736	.0355	29
1.0653	.0316	30

FIGURE 3. Structural return loss.

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Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012
MIL-C-17/196

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2010-034)

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH, TD
Air Force – 19, 99
DLA - IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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